Power Systems S5EL, 2026-27- Class Test -2 (50 minutes, 50 marks)

1. A 3-phase, 25 MVA, 11/66 kV transformer has a leakage reactance of 8% on its own rating. It is to be represented on a common system base of 50 MVA, with the base voltage on the LV side chosen as 11 kV.

(a) Determine the base current and base impedance on the 11-kV side. [4]

(b) Determine the transformer reactance in per-unit on the 50-MVA base and the corresponding actual reactance in ohms referred to the 66-kV side. [4]

(c) Explain why the per-unit impedance of a transformer is the same when referred to either side when the voltage bases are selected in the transformer turns ratio. [2]

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2. The phase currents of an unbalanced three-phase system are given below. Take a = $1\angle 120°$.

$I_a = 100\angle 0° \text{ A}, \quad I_\beta = 80\angle -120° \text{ A}, \quad I_C = 60\angle 120° \text{ A}$

(a) Determine the zero-, positive-, and negative-sequence current components I_0, I_1, and I_2. [4]

(b) Determine the magnitudes of the zero- and negative-sequence currents as percentages of the positive-sequence current. [4]

(c) State the phase order of the positive- and negative-sequence components and identify the sequence component that requires an earth/neutral return path. [2]

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3. At an 11-kV bus of a power system, the Thevenin sequence impedances on a 100-MVA base are as given below. The pre-fault voltage at the bus is $1\angle 0°$ p.u. A solid single-line-to-ground fault occurs on phase a.

$Z_1 = j0.20 \text{ p.u.}, \quad Z_2 = j0.20 \text{ p.u.}, \quad Z_0 = j0.10 \text{ p.u.}$

(a) Determine the sequence currents I_0, I_1, and I_2 and hence the faulted-phase current I_a in per-unit. [4]

(b) Calculate the actual fault current I_a in kA. Also determine the per-unit current for a solid three-phase fault at the same bus. [4]

(c) State how the sequence networks are interconnected for a single-line-to-ground fault and identify the sequence network involved in a balanced three-phase fault. [2]

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4. A string of three identical suspension insulator units is connected between a transmission-line conductor and an earthed tower. Each unit has a self-capacitance C, while the capacitance from each metal junction to earth is 0.10C. The voltage across the complete string is 66 kV. Neglect capacitance between the metal junctions and the line conductor.

(a) Determine the voltage across each of the three insulator units. [4]

(b) Calculate the string efficiency and identify the unit subjected to the maximum voltage stress. [4]

(c) Explain how reducing the shunt capacitance to earth affects the voltage distribution and string efficiency. [2]

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5. A string consisting of four suspension insulator units is to be capacitance graded so that the voltage is equally distributed across all units. The shunt capacitance from each junction to earth is 20% of the capacitance C of the tower-end unit. The total voltage across the string is 80 kV.

(a) Determine the required capacitances C_1, C_2, C_3, and C_4, expressed as multiples of C. [4]

(b) Determine the voltage across each unit and calculate the string efficiency under ideal grading. [4]

(c) Explain why the capacitance is progressively increased towards the line conductor and state one practical disadvantage of capacitance grading. [2]
